

Linear systems and elimination

Optimal uniform row-deletion singular-value constant

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Independently reviewed resolution - 2026-09-11

Affirmative sharp-constant resolution. Theorem 2 and Sections 6-7 prove the sharp constant $c_\theta = \sqrt{h_\theta}$. The upper bound is uniform over all unit-row matrices and even all $m \geq 1$ for sufficiently large n , with squared normalized error $O_\theta(n^{-1/6})$. Theorem 1 supplies matching spherical realizations along every high-aspect-ratio sequence, proving the exact eventual-uniform optimality quantifiers.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects an optimal uniform extremum over matrices and deleted row sets; community impact is a sharp robustness limit for row-sampling methods.

Lean proof and verification evidence - 2026-09-22

The complete original target is Lean verified. The [formal proof at revision c45f9cc](#) proves the sharp Gaussian constant for the actual supremum over every real unit-row matrix. It includes the exact finite bound, a squared error rate uniform over all positive row counts, deterministic near-extremizers along every high-aspect sequence, the supremum limit, and failure of the original eventual-uniform property for every smaller constant. The floor convention, rank deficiency and empty retained selections are included.

Lean formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, with substantial AI-agent assistance. **Matthew J. Colbrook** retains attribution for the original mathematical proof. The original statement, source attribution and dated history below remain preserved.

All [20 required declarations](#) passed authentic LeanCert kernel-trust assertions, the sandboxed Lean4 Comparator and kernel replay on non-root Linux. Each transitive axiom closure contains only *propext*, *Classical.choice* and *Quot.sound*. Exact spectral and analytic arguments establish the sharp variance estimate and asymptotic rate without approximate quadrature or numerical eigenvalue searches. The [operational evidence](#) records immutable inputs, authenticated dependency revisions and successful isolation and rejection controls.

Two nonauthor agents independently reviewed the complete statements and all 47 proof modules: [fidelity review](#) and [correctness review](#). Their separate [fidelity completion review](#) and [correctness completion review](#) audit the Linux evidence. Publication correspondence is reviewed in subsequent addenda. These are independent AI reviews under the repository's adapted Tau Ceti protocol, not external human peer review or official Tau Ceti endorsement. See the [project guide](#) and [formalization metadata](#) for exact scope, review history and reproduction instructions.

Problem statement

Fix $0 < \theta < 1$. For a real $m \times n$ matrix A with every row of Euclidean norm one, put

$$s_\theta(A) = \min_{S \subseteq \{1, \dots, m\}, |S| = \lfloor \theta m \rfloor} \min_{\|x\|_2=1} \|A_S x\|_2, \quad M_{m,n}(\theta) = \sup_{\|a_i\|_2=1} \sqrt{\frac{n}{m}} s_\theta(A).$$

Here a_i is row i of A , and A_S contains exactly the rows indexed by S .

Determine the sharp asymptotic upper constant for $M_{m,n}(\theta)$ as $n \rightarrow \infty$ and $m/n \rightarrow \infty$. In particular, is it

$$c_\theta = \left(\frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2} dt \right)^{1/2}, \quad \mathbb{P}(|G| \leq a_\theta) = \theta, \quad G \sim N(0, 1)?$$

Precisely, the proposed uniform inequality asks whether, for every $\varepsilon > 0$, there exist integers N and R such that

$$M_{m,n}(\theta) \leq c_\theta + \varepsilon \quad \text{whenever } n \geq N \text{ and } m/n \geq R,$$

and whether no smaller constant has this property. The supremum and quantifiers specify the source's uniform $1 + o(1)$ bound; the floor convention handles row-count integrality.

Connection to numerical linear algebra

This asks how well any unit-row linear system can retain its smallest singular value under worst-case row removal, a geometric constraint on robust iterative solvers.

References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, the third question immediately after equation (8). [Published paper](#). [Author preprint](#), v1, §2.3, paragraph following (◇).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1 (background on the same restricted-singular-value heuristic and the different streaming model), §5. [Preprint](#).

Earlier status check — 2026-09-08

Checked the published question and latest arXiv records on 2026-09-08. Targeted searches for “Steinerberger subsingular”, “small subsingular values”, “quantile Kaczmarz optimal constants”, “unit rows smallest singular value”, and 2025/2026 variants found no resolution of the all-matrices extremum. Recent streaming Kaczmarz guarantees do not supply this universal extremal inequality. The random-ensemble limit in the same source is a related but distinct question: it specifies a random construction, whereas this question optimizes over all unit-row matrices. Openness remains subject to the limits of this search.

Audit update — 2026-09-10

Rechecked Steinerberger’s [primary formulation](#), §2.3. Searches for the optimal row-deletion constant and quantile-Kaczmarz follow-ups found no proof of this extremal value. The new [Cai–Chen–Ma–Wu paper](#), SIAM J. Matrix Anal. Appl. **47** (2026), 802–823, proves a tight QRK subsample-size result, not the universal matrix constant asked for here.